

PAUL WANG, MECHANICAL ENGINEERING STUDENT

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WORK EXPERIENCE

Suspension Lead, *Edinburgh University Formula Student (EUFS)* *DEC 2024 – Present*

- Solely responsible for modelling and designing suspension assembly in Simscape & SolidWorks.
- Designed a suspension geometry optimiser from scratch using Simscape multibodies. Created full-car kinematic model, determined optimal suspension geometry and roll axis. Validated quantitative performance benefits of anti-ackermann and anti-dive and used in-house tyre magic formula model.
- FEA & topology optimisation for all suspension and bellcrank components in SolidWorks.
- Represented suspension at Formula Student UK's design presentation, contributed to team's 6th place, topology optimised bellcranks & motion ratio contributed 15 points and mentioned as good design.

Vehicle Dynamics Engineer, *Edinburgh University Formula Student* *SEP 2024 – Present*

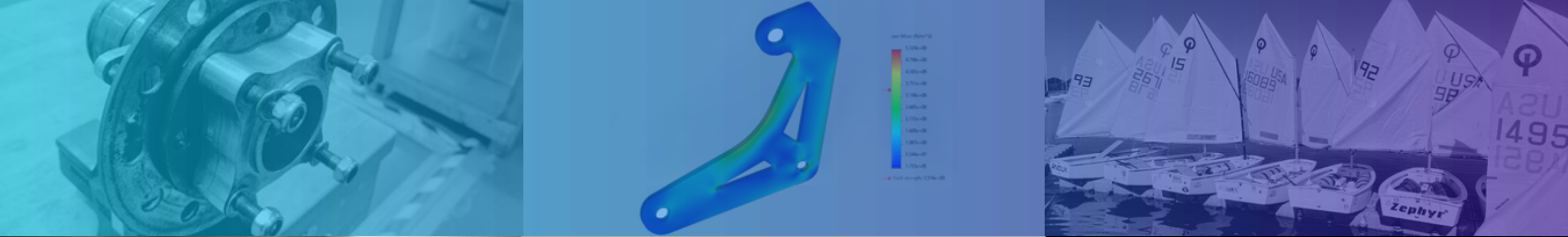
- Manufacturing experience, SolidWorks, additive manufacturing, laser cutting, and hydraulic tools.
- Replaced and repaired brake lines & calipers, suspension components, shocks, and resealed tyres.
- Created a quasi-steady-state point-mass laptime simulator from scratch in MATLAB. Determined the efficacy of large-scale concepts, such as car weight, battery size, & aero targets.
- Advised Aerodynamics team on CFD & aero designs regarding Vehicle Dynamics and overall concept.

Motorboat, Sailboat, and Aircraft Maintenance Technician *JUL 2023 – Present*

- Motorboat & sailboat maintenance at MSC, performed hydraulic steering, drivetrain, and electrical maintenance on motorboats, and fibreglass & carbon fibre composites work for our fleet of sailboats.
- Aircraft repairs, teardowns, and component analysis at TCFT, performed repairs on hydraulics, tyres, brakes, avionics, and fuselage. Performed teardowns on Cessna airframes and engines.

Sailing Instructor, *Minneapolis Sailing Center (MSC)* *MAY 2022 – Present*

- Head coach for the Optimist Racing Team, managed logistics and coached a team of 8-12 students. Solely responsible for designing all lesson plans and tailoring my coaching for each student. Coached Optimist and C420 racing teams at official US Sailing regattas.
- Head instructor for the Adult Laser Racing class, taught complex physics and aeronautical concepts, such as wing-tip vortices, airflow separation, and centre of gravity & roll center dynamics.
- 1500+ work hours, 75+ class sessions taught, named best instructor of 2024 and 2025.



EDUCATION

BEng Honours Mechanical Engineering, University of Edinburgh

SEP 2024 – Present

- Grade: First Class (A2). Completed 10 classes with 160 UK credits.

University Classes Taken for High School, University of Minnesota

SEP 2021 – MAY 2024

- 3.66 Cumulative GPA, 4.0 Technical GPA, 3 Dean's List Honours.
- Completed 17 university classes with 70 US credits. Normal university classes taken full time as secondary school credit. Followed the UMN Mechanical Engineering & Computer Science degree tracks.

Pilot Training, Twin Cities Flight Training (TCFT)

MAY 2022 – JUN 2024

- FAA Licensed Private Pilot. 200+ Landings, 100+ flight hours. Actual instrument experience, ILS landings.

PROJECTS

Software and Programming Languages

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|----------------------------------|---|
| – MATLAB | Differential equations & matrices calculations for Formula Student. |
| – Simulink & Simscape | Simscape multibodies and Simulink for suspension geometry optimiser. |
| – Solidworks | Design, FEA, & topology optimisation for Formula Student. |
| – Java | Created Chess, Minesweeper, Tetris, and hashing tables from scratch. |
| – Python | Created orbital motion simulations and real-time data processors. |
| – HTML & CSS | Used VS code to create my personal website: https://paulhanwang.co.uk . |
| – UNREAL ENGINE 5 | Created 3D renders of the Edinburgh University Formula Student car. |
| – C#, C++, & C | Used Unity Game Engine to create 3D and 2D video games. |

FAA Licensed Private Pilot

- 270+ Landings, 200+ flight hours. Actual instrument experience, ILS landings. 30,000+ miles flown.

Testing and Race Driver

- Testing and Race Driver for Car 128 - Edinburgh University Formula Student

Cinematographer & Graphic Designer

- Created several amateur cinematography films and sailboat racing action videos.
- Created bespoke logo, business card, and website design.
- Used Unreal Engine 5 renders for official Edinburgh University Formula Student marketing posters.

Community Outreach Volunteer

- Volunteered at the Minneapolis Sailing Center as a skipper to give free sailboat rides to members of the public during our Free Sailing Sunday events and local Minneapolis charities during community outreach events.